Performance characterization of PET detectors and systems

Oscilloscope baseline, Pico-TDC performance, and waveform-ML correction

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Why timing matters in TOF-PET

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TOF-PET imaging

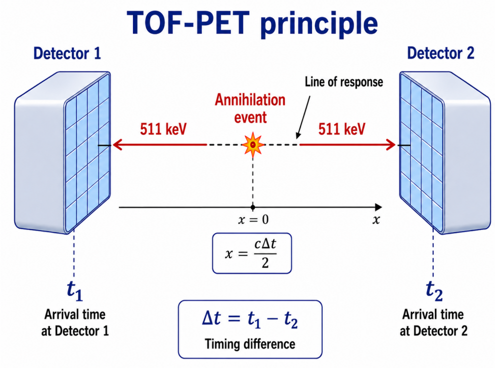
A positron annihilation produces two photons emitted in opposite directions. Two detectors measure the corresponding signals and estimate the **difference in photon arrival time (TOF)**.

Why better timing helps

The arrival-time difference constrains the annihilation position along the line connecting the two detectors:

$$\Delta x = \frac{c\,\Delta t}{2}.$$

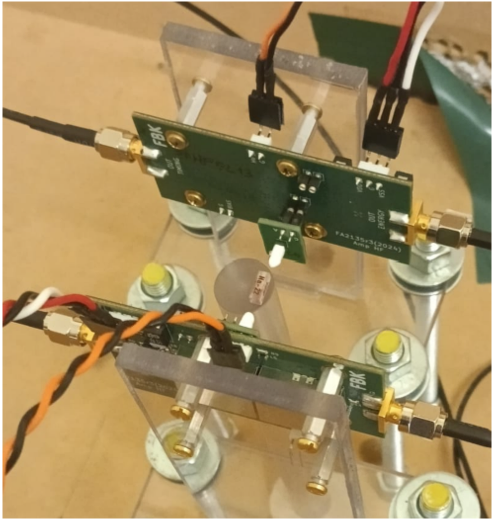
A more precise estimate of Δt means a more precise localization of the event.



Project question

What is the best timing performance achievable, and how can we approach it with a practical and scalable method?

Experimental setup



Detector system

- Two opposing LYSO–SiPM detector modules.
- ^{22}Na positron-emitting source centered between the two detectors.

Principle

A 511-keV photon produces scintillation light in the LYSO crystal, which the SiPM converts into an electrical pulse.

Analog outputs

For both detectors:

- **energy** channel: first stage of amplification. Necessary for event selection, but can also be used for timing.
- **timing** channel: second stage of amplification, specifically designed for timing.

Two ways to acquire the detector signal

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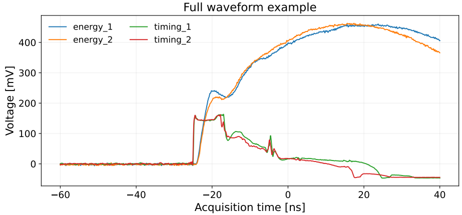
Oscilloscope: Full waveform

The voltage signal is sampled as a function of time.

- complete signal shape acquired;
- flexible timing analysis.



Pico-TDC: Threshold-based

Records the times at which the signal crosses a fixed threshold.

- leading and trailing crossing times;
- Time-over-Threshold (ToT).

```
//************************************************
// File Format Version 3.3
// Janus_5203 Release 3.0.0
// Acquisition Mode: Trigger Matching
// Measurement Mode: Lead Trail
// ToA_LSB = 3.125 ps; ToT_LSB = 3.125 ps; Tstamp_LSB = 12.8 ns
// Run50 start time: Wed Jul  1 21:06:25 2026 UTC
//************************************************
      Tstamp_LSB        TrgID   Brd  Ch E      ToA_LSB      ToT_LSB
         3315422            0    00  01 L       282084            -
                                 00  01 T       306087        24003
                                 00  03 L        65599            -
                                 00  03 T        66039          440
                                 00  03 L       113212          537
                                 00  03 T       113775        48176
```

Same detector signals, two different levels of information: full waveforms versus a small set of timestamps.

Why use the Pico-TDC?

Device

CAEN DT5203: 64-channel picoTDC unit for high-resolution ToA/ToT measurements (3.125 ps LSB).

Practical motivation

- **Lower cost** than high-bandwidth waveform digitization.
- **More channels** available in a compact system.
- Much **smaller data volume** per event.

Main question

Can this readout preserve the timing performance of full waveform acquisition?

Experimental data

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Oscilloscope

- **Bias voltage:** 45–49 V
- **Acquisition:** 100 ns waveforms at 80 GS/s

Bias [V]	Raw	Photopeak
45	20 135	6 763
46	21 510	8 037
47	20 073	8 482
48	20 264	7 049
49	25 054	7 078

Pico-TDC

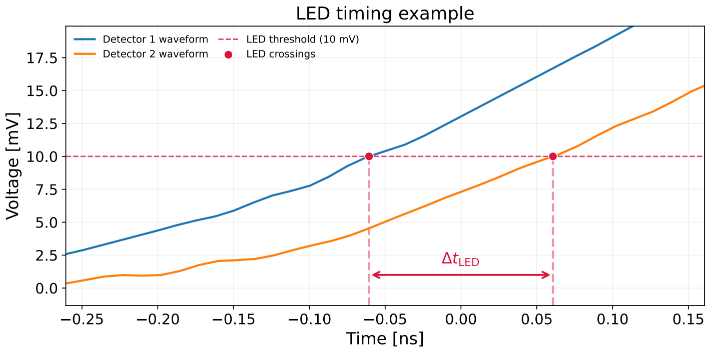
Threshold scan at fixed bias $V = 46$ V:

T_{th} [mV]	Raw	Photopeak
20	49 348	5 793
30	49 688	5 909
40	100 916	11 847
50	50 245	5 931
60	50 271	5 800
70	50 410	5 981

Bias-voltage scan at fixed $T_{\mathrm{th}} = 40$ mV:

Bias [V]	Raw	Photopeak
42	79 127	11 514
43	84 586	11 470
44	90 229	11 554
45	96 008	11 552
46	100 916	11 847
47	104 916	11 881
48	107 613	12 182
49	111 406	12 434

Standard methods for oscilloscope waveforms



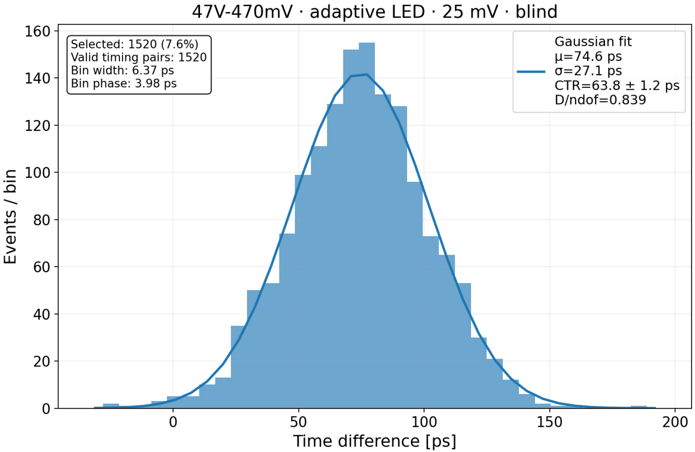
Leading-edge discriminator (LED)

Fixed voltage crossing: simple and robust, but sensitive to amplitude-dependent time walk.

Variant: Constant-fraction discriminator (CFD)

Crossing at a fixed fraction of each pulse amplitude: reduces amplitude-dependent time walk.

CTR extraction



Timing difference

For each selected coincidence event:

$$\Delta t = t_1 - t_2.$$

The selected events form the coincidence time distribution.

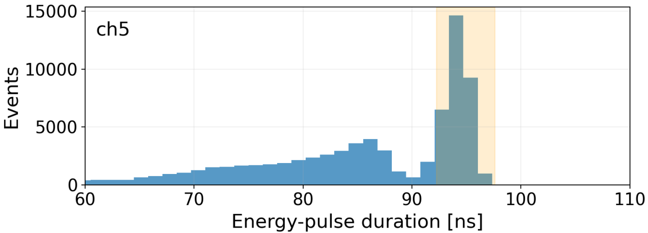
Coincidence Time Resolution (CTR)

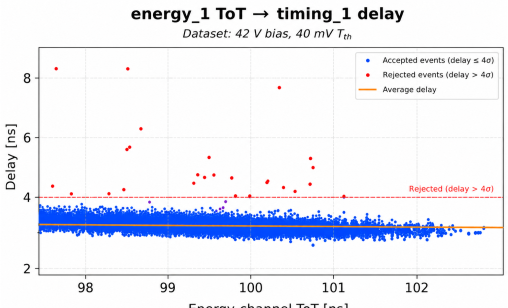
$$\mathrm{CTR} = \mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma \simeq 2.35\,\sigma.$$

Lower CTR means better TOF-PET image quality.

The same Gaussian-fit CTR definition is used for every timing method.

Pico-TDC analysis: from hits to coincidences





Pipeline

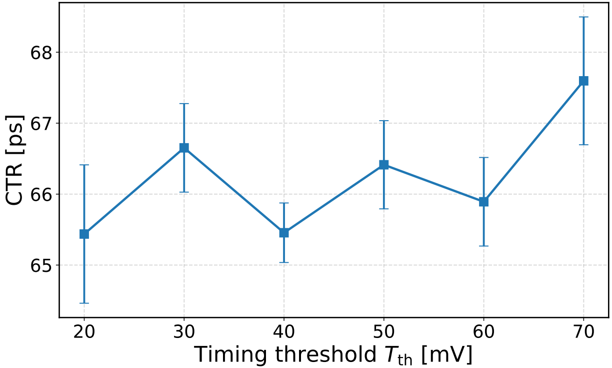
1. Raw timestamp lists
2. Leading/trailing pairs
3. ToT-based photopeak selection
4. Timing-hit matching
5. Mismatch rejection
6. Gaussian CTR fit

Personal contribution

Causal signal matching and filtering strategy to reject mismatched or noisy events while keeping lower timing thresholds usable.

Pico-TDC threshold scan

Threshold scan results with 46 V bias voltage:



Goal

To get the best CTR while preserving detection efficiency.

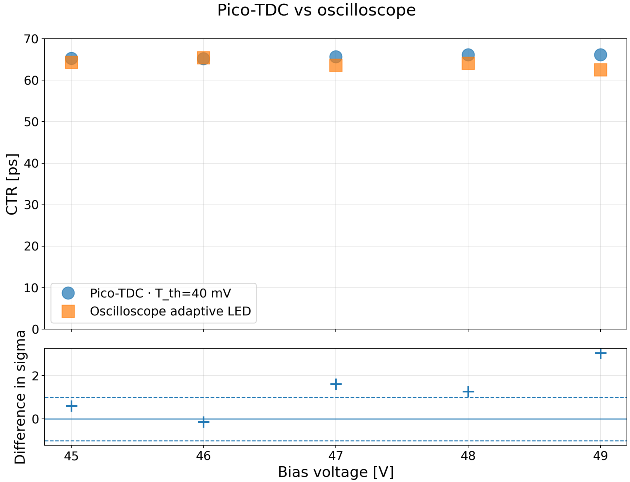
Threshold trade-off

- **Low threshold**: earlier pick-off, but more noise/spurious hits.
- **High threshold**: cleaner crossing, but more time walk and possible efficiency loss.

Analysis result

The threshold can affect timing performance and should be optimized.

Pico-TDC versus oscilloscope reference



Comparison

Error bars show the bootstrap uncertainty of the final fit only. They therefore underestimate the total analysis uncertainty.

Result

Using the timing channels, Pico-TDC can produce **CTR compatible with the oscilloscope reference** (LED method).

Second question: can we do better?

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Idea

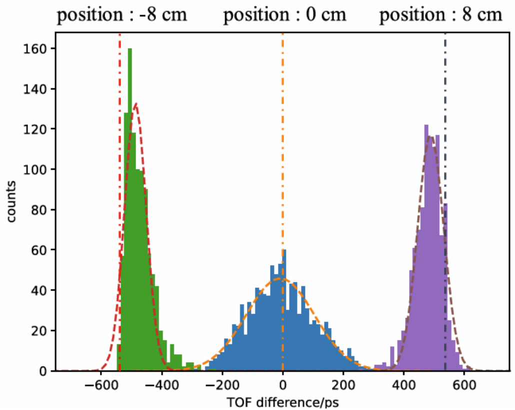
The signal shape contains additional information that standard methods don't use.

Main question

Can a model learn a waveform-dependent timing correction, rather than a black-box TOF prediction?

Risk with Machine Learning

A very flexible model can make the time distribution narrower without necessarily learning the physical timing correction we want.



Source: X. Feng, A. Muhashi, Y. Onishi, R. Ota, and H. Liu, "Transformer-CNN hybrid network for improving PET time of flight prediction," *Physics in Medicine & Biology*, 2024.

What should the model learn?

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Start from standard leading-edge timing

For each coincidence event, the measured time difference is: $\Delta t_{\mathrm{LED}} = t_{\mathrm{LED},1} - t_{\mathrm{LED},2}$.

My modeling proposal

$$\boxed{\Delta t_{\mathrm{LED}} = \underbrace{\mathrm{TOF}}_{\text{true time of flight}} + \underbrace{C_{1,2}}_{\text{calibration bias}} + \underbrace{\delta(s_1, s_2)}_{\text{waveform-dependent error}} + \underbrace{\epsilon}_{\text{random noise}}}$$

Ideal ML target

$$\boxed{\delta(s_1, s_2)}$$

Where (s_1, s_2) is a signal pair. In real applications, the target includes the noise ϵ.

Derived properties

- The correction is independent of TOF.
- Fixed channel offsets are handled by calibration.
- ML is used to correct waveform-dependent timing effects.

Physically constrained correction

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Proposed model architecture: same model for all signals

$$\boxed{y_\theta(s_1, s_2) = g_\theta(s_1) - g_\theta(s_2)}$$

$$\theta^\star = \arg\min_\theta \frac{1}{N}\sum_{i=1}^{N}\left(y_\theta(s_1^i, s_2^i) - y^i_{\mathrm{true}}\right)^2, \qquad y^i_{\mathrm{true}} = \delta(s_1^i, s_2^i) + \varepsilon^i = \Delta t^i_{\mathrm{LED}} - \widehat{C}_{1,2} - TOF$$

Corrected timing estimate

$$\boxed{\widehat{\Delta t} = \Delta t_{\mathrm{LED}} - \widehat{C}_{1,2} - y_\theta(s_1, s_2)}$$

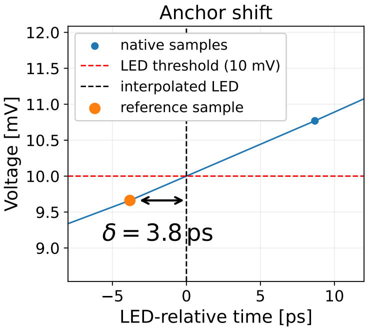
The final output is still a corrected arrival-time difference, so it can be evaluated with the same CTR metric.

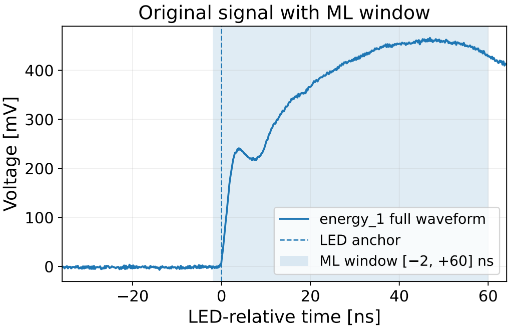
Why this is safer than generic pair-ML

- Same pulse scorer for every detector.
- Learns a timing-error correction, not an absolute position label.
- No need for multiple source positions: a single central source is sufficient.

The proposed formulation gives a translation-invariant, antisymmetric, and more general timing correction.

Windowing and target correction





Windowing

LED time is interpolated, but waveform samples remain on the native grid:

$$\delta = t_{\text{LED}} - t_a$$

Target

Correct for the discrete anchor shift:

$$\boxed{y_{\text{target}} = \Delta t_{\text{LED}} - \Delta\delta - TOF - \widehat{C}_{1,2}}$$

Waveform ML analysis pipeline

Raw waveforms Energy OR Timing Channels → **Preprocessing** selection + LED → **ML dataset** windows + targets → **Model development** train + validation → **Blind evaluation** CTR

Input data

Previous studies mainly considered energy-channel waveforms only. I applied the pipeline also on timing channel data, using the energy channel for the photopeak selection.

Holdout validation

Original dataset → **Development set** + **Blind test**

Development set → **Train** | **Validation**

The **original dataset** is split into a **development set** and a **blind test**. Only the **development set** is used for training and model selection.

Machine learning models

Machine learning models

Model	Description	Scientific question
Linear SVR	Weighted sum of waveform samples	Is the correction mostly linear?
CNN	Learns local waveform patterns	Are nonlinear models better?

SVR: Support Vector Regressor *CNN: Convolutional Neural Network*

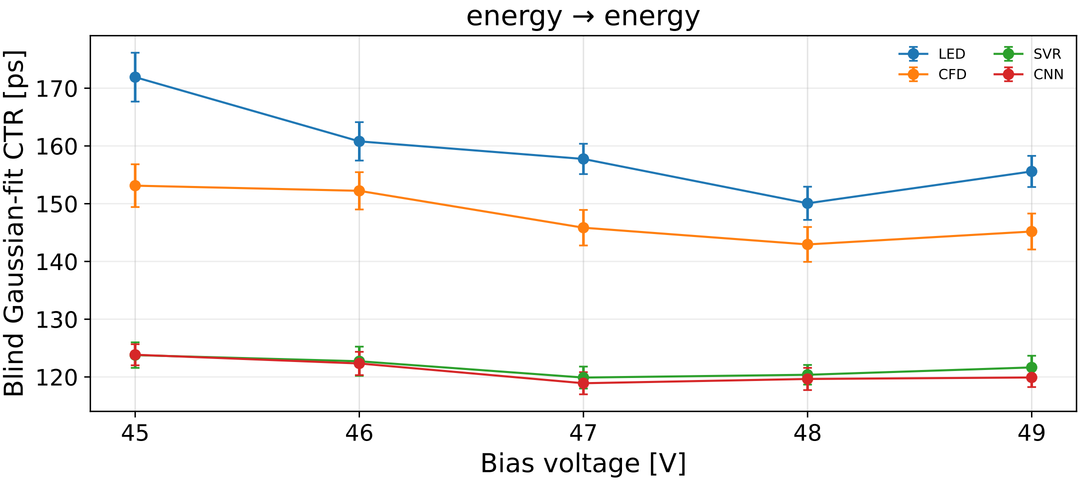
Performance

A model is useful if it reduces CTR on the independent blind set.

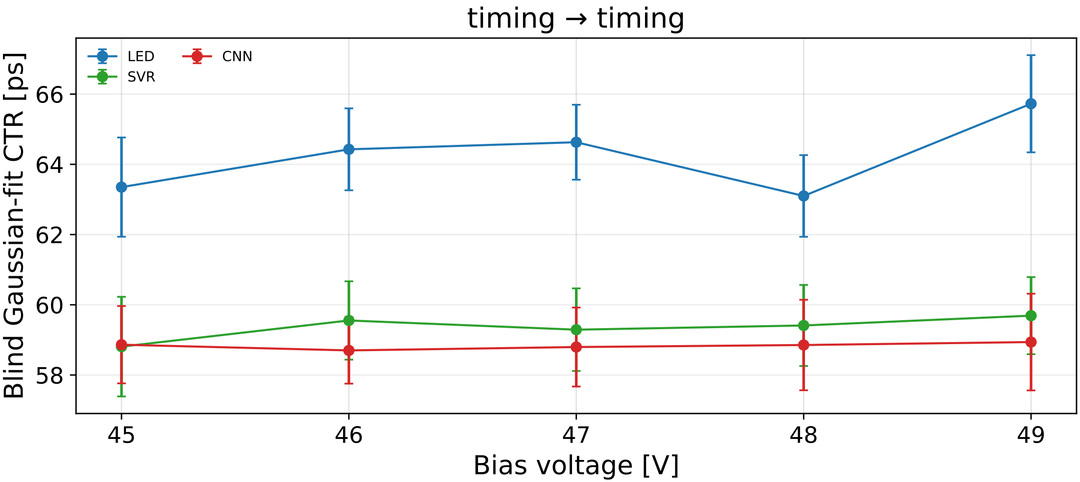
Interpretation

If different models learn similar event-by-event corrections, performance is likely limited by the information in the signal, rather than by model complexity.

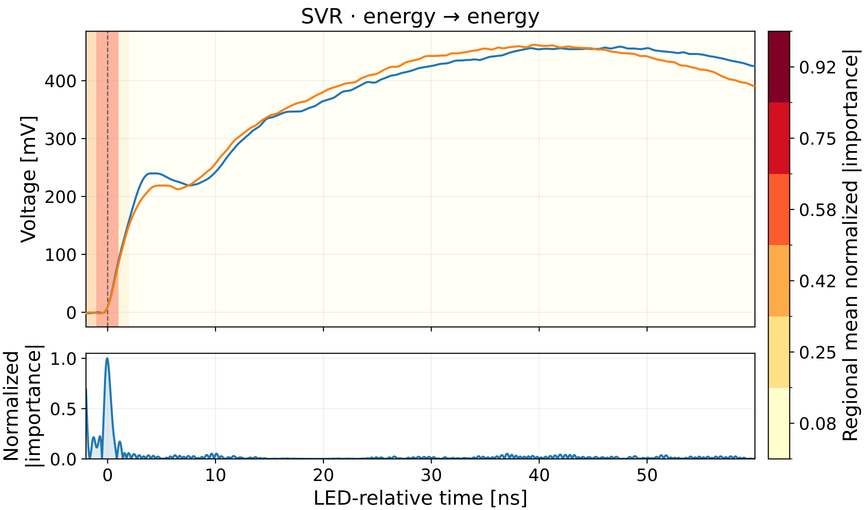
Waveform-ML results: energy-channel timing



Waveform-ML results: timing-channel timing

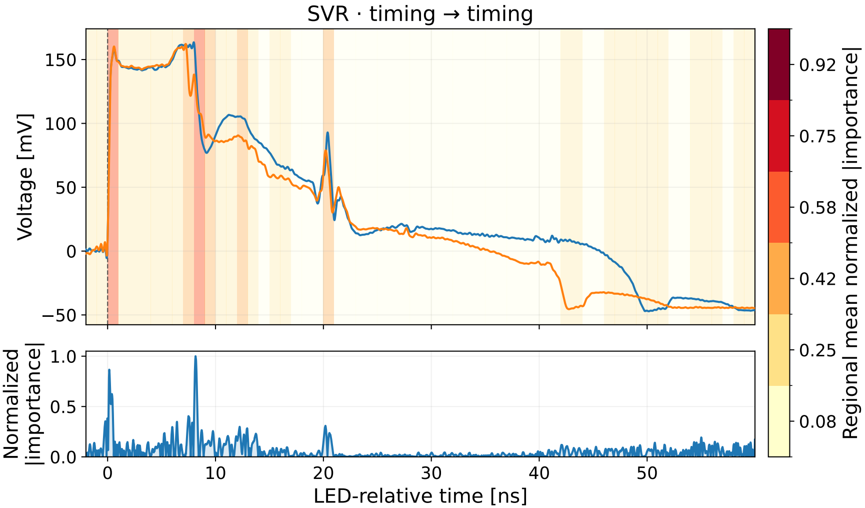


XAI: energy-channel information



The useful timing information is concentrated close to the signal arrival

XAI: timing-channel information



Useful regions remain more than 20 ns after signal arrival.

Conclusions

- Using the dedicated timing channels, Pico-TDC measurements achieve CTR compatible with the oscilloscope reference.
- Machine learning corrects the waveform-dependent timing error, with the largest improvement observed for the energy channels.
- The shared antisymmetric model gives a translation-invariant correction, while a few threshold-crossing times retain much of the useful information.
- Timing-relevant information can persist even nanoseconds after signal arrival.

Machine learning can improve timing and reveal where useful information is encoded in the detector signal.

References and source material

Thank you for your attention.

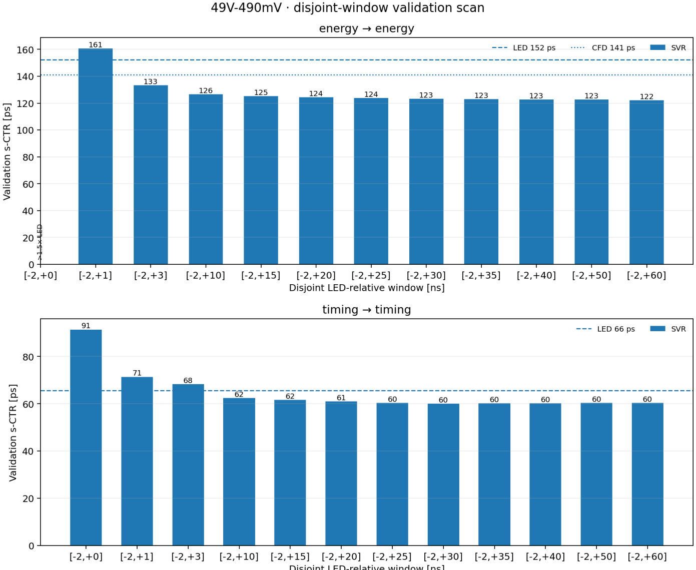
References

- E. Berg and S. R. Cherry, "Using convolutional neural networks to estimate time-of-flight from PET detector waveforms," *Physics in Medicine & Biology*, 2018.
- X. Feng, A. Muhashi, Y. Onishi, R. Ota, and H. Liu, "Transformer-CNN hybrid network for improving PET time of flight prediction," *Physics in Medicine & Biology*, 2024.
- S. Naunheim et al., "Explicit machine-learning corrections and spatial generalization for PET detector timing," *Frontiers in Physics*, 2025.
- L. Y. Chang et al., "A Time-Walk Correction Method for PET Detectors Based on Leading Edge Discriminators," *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2017.

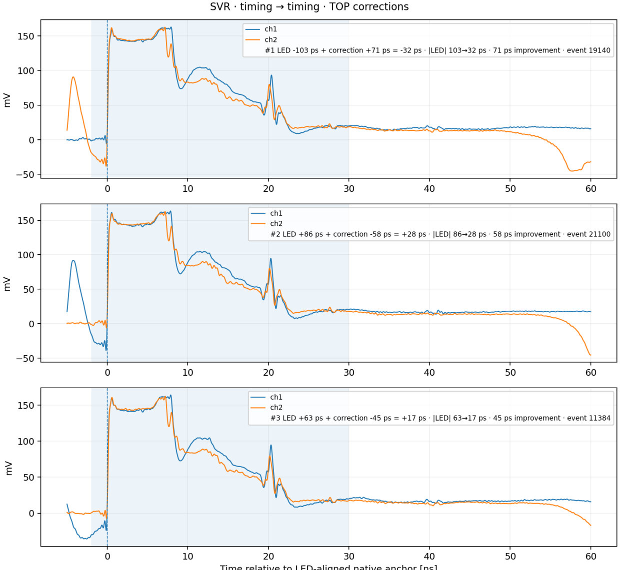
Repository

`tof-pet-timing-uchicago-ONAOSI-prj3`

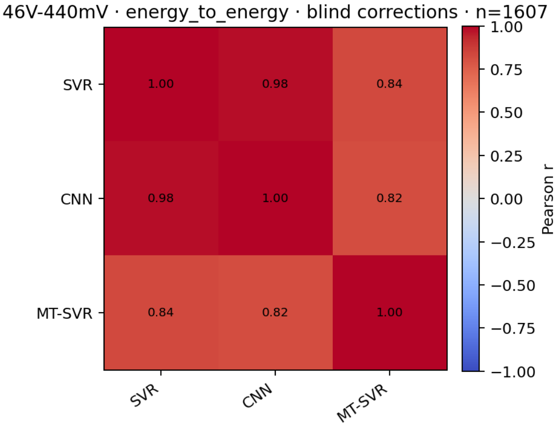
Backup: window scan



Backup: top corrections



Backup: model output correlation



Result

SVR, CNN, and fixed-threshold SVR produce **strongly correlated event-by-event corrections**.

Interpretation

Similar corrections across different models suggest that signal information, not model complexity, limits performance.